

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Backlog Theory Examination March - 2018

PH316 Medicinal Chemistry-I[New Syllabus]

Date: 30/03/2018, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q. 1 Attempt all questions. Each question is of one mark. 20
1. Anticholinergics are also called as
[A]Antispasmodics
[B]Parasympatholytics
[C]Cholinergic Blockers
[D] All of the above
 2. Receptors of Ach are
[A]M₅
[B]N₁
[C]Both A and B
[D] None of the above
 3. Solaneous alkaloids fall in the category of
[A]Antichloinergrics
[B]Adrenergics
[C]Antiemetics
[D] Anti diarrhoeals
 4. Which of the following is a conformational isomer of Ach
[A]Synperiplanar
[B]Synclinal
[C]Eclipsed
[D]Both A and B
 5. Aging is a process that occurs in following class of drugs
[A]Adrenegics
[B] Cholinesterase inhibitors
[C]Purgatives
[D]None of the above
 6. _____ is the precursor of Norepinephrine
[A] Epinephrine
[B] Dopamine
[C] Adrenaline
[D] Nor adrenaline

7. One of the drug is not a α,β - Adrenergic antagoist
[A]Carvidilol
[B]Labetalol
[C]Metoprolol
[D]None of the above
8. Albuterol, Metaproterenol, Tebutaline fall in the following category-
[A] Antitussives
[B] Expectorants
[C]Antiasthmatics
[D] All of the above
9. Antitussive act by
[A]Supperesing cough center
[B] Loosening cough
[C] Breaking mucous
[D] All of the above
10. Purgatives act by
[A] Bulk forming
[B] Osmosis
[C] Emollient action
[D] All of the above
11. Which of the long acting beta bloker used in Glucoma?
[A] Timolol
[B] Levabulol
[C] Cartelol
[D] Betaxolol
12. Activation of the sympathetic nervous system leads to the 'fight and flight' response. Which of these is not part of that process
[A]Vasodilatation in skeletal muscle
[B]Sweating
[C]Bladder relaxation
[D]Increased gut motility
13. Thiophene has_____ pairs of delocalized electrons
[A] 2
[B] 3
[C] 6
[D] 1
14. Furan ring is found in the following drug
[A] Ranitidine
[B] Salbutamol
[C] Propanolol
[D] Physostigmine
15. Histamine is a derivative of:
[A]purine
[B] pyrrole
[C] imidazole
[D] pyridine

16. What is the starting material for the synthesis of Salbutamol?
 [A] Phenyl acetonitrile
 [B] Methyl Selicylate
 [C] 4-Hydroxy propiophenone
 [D] Mesitylene derivatives
17. What is the correct order of reactivity (most reactive first) of pyrrole, furan and thiophene towards electrophiles?
 [A] Furan > Pyrrole > Thiophene
 [B] Pyrrole > Thiophene > Furan
 [C] Pyrrole > Furan > Thiophene
 [D] Thiophene > Pyrrole > Furan
18. Gatterman reaction of Pyrrole provide _____ Product
 [A] 2 and 3 Pyrrole dicarboxylic acid
 [B] 2-Nitro Pyrrole
 [C] Pyrrole 2-aldehyde
 [D] Tetrahydropyrrole
19. Proton pump inhibitors act on
 [A] H₁ receptors
 [B] H₂ receptors
 [C] M₁ receptors
 [D] G receptors
20. A Pro-drug is _____
 [A] Inactive outside but active inside
 [B] Active outside but Inactive inside
 [C] Both A and B
 [D] None of the above

SECTION – II

- Q 2** Attempt any **TWO** of the following
- | | | |
|----------|--|-----------|
| A | Classify Muscarinic drugs and discuss stereochemistry of Acetylcholine. | 05 |
| B | Give SAR of Sympathomimetics with special reference to Phenylethanolamine. Give synthesis of Salbutamol. | 05 |
| C | Classify and mechanism of action of Neuromuscular Blocking Agents. Give synthesis of Propranolol. | 05 |
- Q 3** Attempt any **TWO** of the following
- | | | |
|----------|--|-----------|
| A | Classify and give description of any two drugs of each class of Prokinetics. | 05 |
| B | Classify and give mechanism of actions of Laxatives. | 05 |
| C | Give Mechanism of action of antiallergic drugs. Write synthesis of Chlorpheniramine. | 05 |

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Write structure and mechanism of G Protein-coupled receptor (GPCR). **05**
- B** Give **any TWO** methods of preparation of the following:(**ANY TWO**) **05**
(1)Imidazole (2) Thiophen (3) Furan
- C** Write classification, SAR and chemical mechanism of action of Proton Pump Inhibitors. **05**
- D** Classify H₁ receptor antagonist with suitable examples. **05**
- E** Write reaction and mechanism of Hantzsch synthesis for Pyrrole. **05**
- F** **Write a comments on:** **05**
1. Impurities of thiophene in benzene can be removed by cold sulphuric acid.
 2. Electrophilic substitution reaction in furan occurs at 2nd and 5th position.
- Q 5** Attempt any **FOUR** of the following
- A** Give **any TWO** reactions of the following:(**ANY TWO**) **05**
(1) Pyrrole (2) Pyridine (3) Oxazole
- B** Draw the structure of: **05**
(1) Quinolene (2) Triazole (3) Pyrazole (4) Indole (5) Pepiridine
- C** Define and classify prodrugs with suitable examples. Enumerate various applications of prodrug. **05**
- D** Classify antialergic drugs with suitable examples. **05**
- E** What is CYP450? Give its function and importance. **05**
- F** **Write a comments on:** **05**
1. Proton pump inhibitors act as pro-drug.
 2. Aniline is more basic than Pyridine.
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